

System Installation Pre-requisites

System Requirements and Setup Environment Instructions

1. Operating System

- Windows 10/11 (Standard environment used for development and verification).
- Alternative: macOS (10.15+) or Ubuntu (20.04+).

2. Command Line Tools

- **Git**: Essential for cloning the repository and managing versions. Download from git-scm.com.

3. Package & Environment Managers

- **Anaconda** (Recommended): Provides robust environment isolation and pre-compiled mathematical binaries.
- **Python (Standard)**: Standard installation with pip and venv.

4. Integrated Development Environment (IDE)

- **VS Code**: Visual Studio Code editor recommended.

5. Library Specifications

Ensure the libraries defined in the project's root requirements.txt are installed to execute python modeling:

- NumPy ($\geq 1.22.0$)
- Pandas ($\geq 1.4.0$)
- Matplotlib ($\geq 3.5.0$) and Seaborn ($\geq 0.11.0$)
- Scikit-Learn ($\geq 1.0.0$)
- Joblib ($\geq 1.1.0$)
- Flask ($\geq 2.0.0$)